

Module 2: Conditional and Logical Operations

2.1 Concept

Conditional and logical functions in DAX let you **make decisions** based on conditions. They are commonly used to:

- **Categorize transactions** (e.g., High/Low Sales).
- **Create flags** (e.g., Discounted vs No Discount).
- **Handle multiple conditions** (e.g., IF + AND/OR).

👉 These functions make reports **smarter and easier to interpret**.

2.2 Key Functions Overview

Function	What it Does	Example (on dataset)
IF(condition, true, false)	Simple branching	IF(Sales[Discount] > 0, "Discounted", "No Discount")
SWITCH(expression, ...)	Handles multiple conditions	Classify Sales into High, Medium, Low
AND(cond1, cond2)	Both must be TRUE	AND(Sales[Quantity] > 2, Sales[Discount] > 0.1)
OR(cond1, cond2)	At least one must be TRUE	OR(Sales[Quantity] > 4, Sales[Discount] > 0)
NOT(cond)	Opposite of condition	NOT(Sales[Discount] = 0)

2.3 Practical Examples

Example 1: Flagging Discounted Sales

```
Discount Flag :=  
IF(  
    Sales[Discount] > 0,  
    "Discounted",  
    "No Discount"  
)
```

- ◆ Creates a Yes/No style flag.

Example 2: Classifying Sales by Value

```
Sales Value Band :=  
SWITCH(  
    TRUE(),  
    Sales[SalesAmount] > 5000, "High",  
    Sales[SalesAmount] > 2000, "Medium",  
    "Low"  
)
```

- ◆ Categorizes transactions into sales tiers.

Example 3: Fast Shipping Indicator

```
Fast Shipping :=  
IF(  
    DATEDIFF(Sales[OrderDate], Sales[ShipDate], DAY) <= 3,  
    "Yes",  
    "No"  
)
```

- ◆ Flags orders that ship within 3 days.

Example 4: Complex Condition with OR

```
High Priority Order :=
IF(
    OR(Sales[Quantity] > 3, Sales[Discount] >= 0.1),
    "High Priority",
    "Normal"
)
```

- ◆ Labels bulk or highly discounted orders as "High Priority".

2.4 Exercises (Hands-On)

1. Shipping Risk Flag

Create a column that returns "Risky" if:

- Discount > 0.1 **AND** Shipping Days > 7, otherwise "Safe" .

2. Profit Category

Create a calculated column:

- Profit > 2000 → "Very Profitable"
- Profit > 1000 → "Profitable"
- Else → "Low Profit"

3. Discount Category (using SWITCH)

- 0 → "No Discount"
- <= 0.1 → "Low Discount"
- 0.1 → "High Discount"

2.5 Visualizations in Power BI

1. Clustered Column Chart

- X-axis = Discount Category
- Y-axis = Total Sales

2. Table/Matrix

- Product | Sales Value Band | Profit Category

3. Pie Chart

- Share of "Fast Shipping" vs "Slow Shipping"

2.6 Exercise Answers (For Trainers)

1. Shipping Risk Flag

```
Shipping Risk :=  
IF(  
    AND(Sales[Discount] > 0.1, DATEDIFF(Sales[OrderDate], Sales[ShipDate], DAY) > 7),  
    "Risky",  
    "Safe"  
)
```

- Expect very few "Risky" rows, since discounts + long delays are rare in the dataset.

2. Profit Category

```
Profit Category :=  
SWITCH(  
    TRUE(),  
    Sales[Profit] > 2000, "Very Profitable",  
    Sales[Profit] > 1000, "Profitable",  
    "Low Profit"  
)
```

- "Very Profitable" sales come mostly from **Electronics** (high-priced items).
- "Low Profit" appears mostly in **Fashion** and some Accessories.

3. Discount Category

```
Discount Category :=  
SWITCH(  
    TRUE(),  
    Sales[Discount] = 0, "No Discount",  
    Sales[Discount] <= 0.1, "Low Discount",  
    "High Discount"  
)
```

- Dataset was generated with random discounts of **0**, **5%**, **10%**, **15%**, so you'll see a nice distribution across all three.

✓ By the end of Module 2, students will:

- Use `IF` and `SWITCH` to create categories.
- Apply logical operators (`AND` , `OR` , `NOT`) in conditions.
- Build **flags and risk categories** directly from raw dataset columns.